<u>Report containing the Summary , analysis and interpretations after conducting analysis</u>

This is a report containing the important findings and inferences I have drawn after conducting data analysis for a dataset containing macroeconomic indicators that were tracked over a period of time [but I have only used the latest year data for analysis for ease of comparing and getting a view of how all of the countries are doing for the most current year data available(2021 in this dataset).]

Details regarding the dataset:- It provides annual macroeconomic indicators for global economies. Each observation captures country-level performance for a given year, covering key metrics across aggregate production, national income, international trade, consumption, gross investment, demographics, and sectoral output .

Column/ Field headings	Description [what they indicate]
CountryID	Unique identifier assigned to each country.
Country	Name of the country being observed.
Year	Year corresponding to the macroeconomic data.
AMA Exchange Rate	Exchange rate reported using the Alternative Monetary Arrangement methodology.
IMF Based Exchange Rate	Exchange rate reported according to IMF standards.
Population	Total population of the country.
Currency	Official currency used by the country.
Per Capita GNI	Gross National Income per person; commonly used to classify countries by income level.
Agriculture, Hunting, Forestry and Fishing (ISIC A–B)	Economic output generated from primary sector activities.[Natural resources extracted or taken from earth]
Changes in Inventories	Increase or decrease in inventories held by producers during the year.
Construction (ISIC F)	Value added by the construction sector.
Exports of Goods and Services	Total value of goods and services sold to foreign countries.

Final Consumption Expenditure	Total spending on final goods and services within the economy.
General Government Final Consumption Expenditure	Government expenditure on public goods and services.
Gross Capital Formation	Total investment made within the economy, including fixed assets and inventory changes.
Gross Fixed Capital Formation	Investment in fixed assets such as machinery, infrastructure and buildings.
Household Consumption Expenditure	Consumer spending by households on goods and services.
Imports of Goods and Services	Total value of goods and services purchased from other countries.
Manufacturing (ISIC D)	Value added generated by manufacturing industries.
Mining, Manufacturing and Utilities (ISIC C–E)	Combined industrial sector output.
Other Activities (ISIC J–P)	Output generated from service sectors including finance, education, ICT and public administration.
Total Value Added	Total economic value created across all industries before taxes and subsidies.
Transport, Storage and Communication (ISIC I)	Value added by transportation, logistics and communication sectors.
Wholesale, Retail Trade, Restaurants and Hotels (ISIC G–H)	Output generated by commercial and hospitality industries.
Gross National Income (USD)	Total income earned by residents of a country, including income from abroad.
Gross Domestic Product (GDP)	Total market value of all final goods and services produced within a country's borders.

Description of Python Files

1. clean-data.py

Purpose

This script performs the initial data cleaning and preprocessing required before any analysis is conducted. It prepares the raw dataset for further statistical analysis and feature engineering.

Operations Performed

- Loads and explores the dataset structure.
- Standardizes column names by converting them to lowercase and replacing spaces with underscores.
- Checks missing values and missing value percentages.
- Identifies and removes duplicate observations.
- Converts numeric columns into appropriate numeric data types.
- Detects outliers using the Interquartile Range (IQR) method.
- Automatically identifies the GNI column.
- Classifies countries into:
    - Low Income
    - Lower Middle Income
    - Upper Middle Income
    - High Income
- Creates a **Development Status** variable:
    - Under-developed
    - Developing
    - Developed
- Extracts only the latest available observation (2021) for each country and exports the cleaned dataset. [As there are some countries that were part of other countries or that were not together or joint as one country, for example, Russia was known as USSR which consisted of a large number of countries that then broke off or split from it , similarly sometime back Germany was split into two countries : East Germany and West Germany] - Taking the most recent year allows ease of comparison while conducting analysis - to see how each of the countries are doing with respect to the different macroeconomic indicators they have been assessed against, allowing ease of spotting current correlations or trends that are occurring. [The most recent data in this dataset is for the year 2021 - so , have filtered out and cleaned the data to include only those records where the year is equal to 2021]

2. feature-engg-1.py

Purpose

This script creates new macroeconomic indicators from the cleaned dataset to provide more meaningful measures of economic performance. These engineered variables are later used in descriptive, univariate, bivariate and multivariate analyses.

Features Created

The script generates variables including:

- Export Dependency
- Import Dependency
- Trade Openness
- Investment Ratio
- Government Share
- Household Consumption Share
- Manufacturing Share
- Agriculture Share
- Services Share
- GDP per Person
- Investment per Person
- Exports per Person
- Imports per Person

It also:

- Creates quartile-based economic groups (Low, Medium, High, Very High)
- Validates engineered variables
- Produces summary statistics
- Exports the feature-engineered dataset

3. descriptive_stat.py

Purpose

This script provides descriptive statistics to summarize the characteristics of every numerical variable before advanced analysis is performed.

Analysis Performed

For every numerical variable it calculates:

- Mean

- Median
- Standard deviation
- Minimum
- Maximum
- Range
- Interquartile Range (IQR)
- Coefficient of Variation
- Skewness
- Kurtosis
- Missing values

Visualizations

The script generates:

- Histograms
- Density plots
- Boxplots

These visualizations help understand the distribution, variability and presence of outliers in macroeconomic indicators.

4. univariate-analysis.py

Purpose

This script examines one macroeconomic variable at a time to understand its statistical distribution and identify unusual observations.

Analysis Performed

For each numeric macroeconomic variable, the script computes:

- Mean
- Median
- Standard deviation
- Minimum and maximum values
- Skewness
- Kurtosis
- Quartiles
- IQR boundaries
- Number of outliers

Outlier Analysis

Countries with unusually high or low values are identified using the 1.5 × IQR rule.

Visualizations

For every variable the script produces:

- Histogram with Kernel Density Estimate (KDE)
- Mean and median reference lines
- Boxplot for outlier detection

Variables analysed

All macroeconomic numerical variables individually, including GDP, GNI, population, exports, imports, investment, sectoral outputs and engineered indicators.

5. bivariate-analysis1.py

Purpose

This script investigates relationships between pairs of macroeconomic variables to understand how one economic indicator changes with another.

Analysis Performed

- Pearson correlation matrix
- GDP correlation ranking
- GNI correlation ranking
- Scatter plots
- Linear regression trend lines
- Pairplot matrix

Variable Comparisons

The script compares relationships including:

Variables Compared	Purpose
GDP vs GNI	Relationship between production and national income
GDP vs Population	Effect of country size on economic output

GDP vs Exports	Export-led growth relationship
GDP vs Imports	Relationship between imports and economic activity
GDP vs Gross Capital Formation	Investment and economic growth
GDP vs Trade Openness	Effect of international trade on GDP
Investment Ratio vs GDP Growth	Association between investment intensity and economic growth

6. multivariate_analysis1.py

Purpose

This script studies how multiple macroeconomic variables jointly influence economic performance and identifies broader structural patterns across countries.

Analysis Performed

Multiple Linear Regression

Predicts GDP using several explanatory variables simultaneously.

Independent Variables

- Population
- Exports
- Imports
- Gross Capital Formation
- Trade Openness
- Investment Ratio
- Manufacturing Share
- Services Share
- Agriculture Share

Dependent Variable

- Gross Domestic Product (GDP)

Outputs include:

- Regression coefficients
- Predicted GDP values
- Residuals

- R^2
- RMSE
- Feature importance plots
- Actual vs predicted scatter plots

Principal Component Analysis (PCA)

Reduces many macroeconomic indicators into a smaller number of principal components to identify the dominant sources of variation among countries.

Variables include:

- GDP
- GNI
- Population
- Exports
- Imports
- Investment
- Trade Openness
- Investment Ratio
- Manufacturing Share
- Agriculture Share
- Services Share
- Government Share
- Household Share
- GDP per Person

Outputs include:

- Scree plot
- PCA loadings
- PCA scores
- PCA biplot

K-Means Clustering

Groups countries with similar macroeconomic characteristics into clusters to identify structural similarities and development patterns.

7. group_comparisions.py

Purpose

This script compares macroeconomic indicators across different categories of countries to identify differences in economic structure and development.

Groups Compared

- Development Status
- Income Group
- GDP Group
- Population Group

Variables Compared

- Trade Openness
- Investment Ratio
- Government Share
- Household Share
- Manufacturing Share
- Agriculture Share
- Services Share
- Export Dependency
- Import Dependency

Analysis Performed

For each group, the script computes:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum

Visualizations

- Bar charts
- Boxplots
- Line charts

The script also exports group averages and automatically identifies which groups have the highest and lowest average values for each macroeconomic indicator.

Overall Analytical Workflow

The complete project follows a structured macroeconomic data analysis pipeline:

1. **Data Cleaning:** Cleans the raw dataset, standardizes variables, removes duplicates, detects outliers, and classifies countries into income and development groups.
2. **Feature Engineering:** Creates derived macroeconomic indicators such as trade openness, investment ratios, sectoral shares, and per capita measures.
3. **Descriptive Statistics:** Summarizes the central tendency, dispersion, distribution, and missing values of all numerical variables.
4. **Univariate Analysis:** Examines the distribution and outliers of individual macroeconomic variables.
5. **Bivariate Analysis:** Investigates pairwise relationships between key macroeconomic indicators using correlations and regression-based visualizations.
6. **Multivariate Analysis:** Explores simultaneous relationships among multiple variables using multiple regression, Principal Component Analysis (PCA), and K-Means clustering.
7. **Group Comparison:** Compares macroeconomic performance across income groups, development status, GDP groups, and population groups using statistical summaries and comparative visualizations.

Analysis of the Graphs generated from univariate, bivariate and multivariate analysis conducted using python

Univariate Graphs:-

Distribution graphs inference or insights

Metric	**Distribution Shape**	**Central Tendency (Mean vs. Median)**	**Primary What It Indicates**	**Key Analytical Inferences**
Exports Per Person	Highly Right-Skewed	**Mean: 11,865.75**	Extreme concentration of export volume in top-tier economies.	1. Heavy global export inequality, where a few high-volume nations pull the mean far above the median.

		Median: 2,251.41		2. Requires logarithmic transformation (log(X)) prior to linear modeling.
Imports Per Person	Highly Right-Skewed	**Mean: 10,837.73** **Median: 2,795.95**	Heavy purchasing power concentration in wealthy countries.	1. Import demand is heavily skewed toward high-income nations, with most nations buying at much lower volumes. 2. Requires log-transformation for parametric analysis.
GDP Per Person	Highly Right-Skewed	**Mean: 18,480.27** **Median: 6,776.03**	Severe global wealth and productivity disparity.	1. Reflects the standard global income gap; most countries cluster at lower GDP per capita levels. 2. High degree of positive skewness necessitates log-scaling.
Investment Per Person	Highly Right-Skewed	**Mean: 4,341.19** **Median: 1,571.47**	Capital deployment is heavily concentrated in rich nations.	1. Per-capita capital accumulation is minimal in most nations but reaches massive peaks in high-income economies. 2. Outliers distort standard linear relationships.
Agriculture Share	Highly Right-Skewed	**Mean: 0.11** **Median: 0.07**	Structural shift away from agricultural reliance globally.	1. Most economies maintain low reliance on agriculture (<7% median), while a long tail of developing nations retains high reliance (>30%). 2. Serves as a strong proxy for structural economic maturity.
Government Share	Moderately Right-Skewed	**Mean: 0.19**	Stable state spending across	1. State spending relative to GDP stays tightly bound around 15% to

		Median: 0.17	most economies with niche outliers.	20% globally. 2. Upper outliers represent extreme state-led or highly centralized economies.
Trade Openness	Moderately Right-Skewed	**Mean: 0.93** **Median: 0.80**	Baseline international trade integration with extreme hub outliers.	1. Typical countries trade roughly 80% --90% of their GDP. 2. Values >3.0 highlight small, specialized global trade/financial hubs (e.g., Singapore, Luxembourg).
Services Share	Approximately Normal	**Mean: 0.34** **Median: 0.33**	Symmetric, bell-shaped distribution across global economies.	1. Service sector share displays strong structural balance across countries, centering around 33%. 2. Safe to use directly in parametric models without non-linear transformation.
Household Share	Approximately Normal	**Mean: 0.63** **Median: 0.62**	Highly consistent global consumption baseline.	1. Macroeconomic consumption remains globally stable, taking up 62%--63% of GDP across varying income levels. 2. Ready for direct inclusion in linear regression models.
Investment Ratio	Approximately Normal	**Mean: 0.24** **Median: 0.23**	Unimodal, bell-shaped capital formation baseline.	1. Economies consistently channel 23% --24% of their total output back into gross capital formation. 2. Low variance and lack of severe outliers make it ideal for modeling in raw form.

Trade Openness	Moderate-High Outlier Intensity Range: ~0.1 to ~4.0	**Median: 0.80** **Mean: 0.93** **IQR Span:** ~0.45 to ~1.10 (Right-skewed)	Measures the total volume of trade (Exports + Imports) as a proportion of overall economic output (GDP).	High positive skew with extreme outliers (>3.0) representing highly trade-integrated or re-export heavy economies (e.g., trade hubs). The central majority of economies cluster between 0.4 and 1.1.
Export Dependency	**Moderate-High Outlier Intensity** Range: ~0.05 to ~2.1	**Median:** 0.35 **Mean:** 0.43 **IQR Span:** ~0.20 to ~0.55 (Right-skewed)	Measures reliance on export revenues relative to total national GDP.	Strongly right-skewed with a sharp concentration around 0.25–0.45. Outliers exceeding 1.0 highlight small, highly specialized, or export-driven economies (e.g., resource exporters/financial hubs).
Import Dependency	Moderate Outlier Intensity Range: ~0.05 to ~2.0	Median: 0.43 Mean: 0.50 IQR Span: ~0.25 to ~0.60 (Right-skewed)	Measures the extent to which domestic consumption and production rely on foreign goods and services.	Slightly higher baseline than Export Dependency (Median 0.43 vs 0.35). Moderate right tail demonstrates that most nations maintain modest import ratios, while a few small, non-resource-rich economies import significantly more than their domestic output.

Gross Domestic Product (GDP)	Extreme Outlier Intensity Range: ~0 to ~2.3 × 10^{13}	Median: 2.80 × 10^{10} (28B) Mean: 4.52 × 10^{11} (452B)	Represents the total monetary value of all finished goods and services produced within a country.	Massive structural inequality in global economic output. The mean is over 16x larger than the median, heavily skewed by a tiny group of mega-economies (e.g., USA, China) in the long right tail (> 15–20 Trillion).
Gross National Income (GNI)	**Extreme Outlier Intensity** Range: ~0 to ~2.3 × 10^{13}	**Median:** 2.61 × 10^{10} (26.1B) **Mean:** 4.52 × 10^{11} (452B)	Measures total domestic and foreign value created by a country's residents (GDP + Net Foreign Factor Income).	Mirrors GDP distribution almost identically with extreme right skewness. Demonstrates that global wealth/income distribution closely aligns with geographic economic output scale, heavily dominated by top-tier global powers.
Wholesale, Retail Trade, Restaurants & Hotels	**Extreme Outlier Intensity** Range: ~0 to ~3.5 × 10^{12}	**Median:** 3.89 × 10^{9} (3.89B) **Mean:** 5.93 × 10^{10} (59.3B)	Tracks value added from consumer-facing trade, retail, service, and tourism sectors.	Highly concentrated in a few domestic consumer superpowers (right tail extending past 3 Trillion). The vast majority of economies fall into the first bin (< 200B), emphasizing that service scale directly tracks total population and market size.
Transport, Storage & Communication	**Extreme Outlier Intensity** Range: ~0 to ~2.5 × 10^{12}	**Median:** 2.05 × 10^{9} (2.05B) **Mean:** 3.99 × 10^{10} (39.9B)	Reflects the scale of logistics, national transport networks, and telecommunications infrastructure.	Severe right-skewed distribution. High values (> 2 Trillion) belong to continental-scale infrastructure markets. The mean is nearly **20x the median**, showing that foundational logistics scale rapidly only in large industrial/tech hubs.

Total Value Added	**Extreme Outlier Intensity** Range: ~ 0 to ~2.3×10^{13}	**Median:** 2.49×10^{10} (24.9B) **Mean:** 4.32×10^{11} (432B)	Represents net economic output across all industries combined (GDP minus intermediate taxes/subsidies).	Heavily correlates with the overall GDP distribution pattern. The extreme right-tail outlier gap highlights that the top percentile of nations generates a disproportionate share of global gross added value.
Other Activities	**Extreme Outlier Intensity** Range: ~0 to ~1.25×10^{13}	**Median:** 8.63×10^{9} (8.63B) **Mean:** 1.91×10^{11} (191B)	Covers public administration, defense, education, healthcare, real estate, and personal services.	Significant rightward bias driven by large government expenditure and massive domestic service sectors in high-GDP countries. Shows a median-to-mean skew factor of **>22x**, the highest disparity among major economic sub-sectors.
Mining, Manufacturing & Utilities	**Extreme Outlier Intensity** Range: ~0 to ~5.8×10^{12}	**Median:** 5.97×10^{9} (5.97B) **Mean:** 9.75×10^{10} (97.5B)	Measures secondary sector output (industrial production, raw material extraction, power generation).	Skewed heavily by global manufacturing powerhouses (tail reaching nearly 6 Trillion). While most developing/small nations have minimal industrial output (< 100B), major industrial centers pull the mean significantly above the median.

Imports of Goods and Services	**Extreme Outlier Intensity** Range: ~0 to ~3.4×10^{12}	**Median:** 1.13×10^{10} (11.3B) **Mean:** 1.26×10^{11} (126B)	Measures the total monetary value of foreign goods and services purchased by domestic buyers.	Severe right skew where the mean is ~11x higher than the median. Over 160 economies cluster below 100B, while global importing powerhouses extend the tail past 3 Trillion.
Exports of Goods and Services	**Extreme Outlier Intensity** Range: ~0 to ~3.5×10^{12}	**Median:** 1.00×10^{10} (10.0B) **Mean:** 1.30×10^{11} (130B)	Measures total domestic value created from selling goods and services to foreign nations.	Strong structural symmetry with import absolute scale. Mean is **13x the median**, highlighting extreme concentration among major export nations (top tail reaching ~3.5T).

Manufacturing	**Extreme Outlier Intensity** Range: ~0 to ~4.9×10^{12}	**Median:** 3.28×10^9 (3.28B) **Mean:** 7.28×10^{10} (72.8B)	Represents net added value specifically from physical product transformation and industrial manufacturing.	Heavy concentration at lower values, with a mean-to-median ratio of **~22x**. Indicates that industrial production capability is concentrated in a tiny fraction of global manufacturing powerhouses.
Construction	**Extreme Outlier Intensity** Range: ~0 to ~1.25×10^{12}	**Median:** 1.41×10^9 (1.41B) **Mean:** 2.50×10^{10} (25.0B)	Tracks value added from building infrastructure, residential property, and civil engineering projects.	Nearly 190 countries fall into the baseline bin (< 50B). Extreme right outliers (~1.2T) reflect major infrastructure booms and urbanization scales in high-population economies.

Household Consumption Expenditure	**Extreme Outlier Intensity** Range: ~0 to ~1.6×10^{13}	**Median:** 1.68×10^{10} (16.8B) **Mean:** 2.50×10^{11} (250B)	Measures total private household spending on consumer goods and services.	Highly unequal distribution across nations. Mean is **~15x higher** than the median, strongly pulled rightward by massive domestic consumer markets (reaching up to 16 Trillion).
General Government Final Consumption Expenditure	**Extreme Outlier Intensity** Range: ~0 to ~3.4×10^{12}	**Median:** 4.30×10^{9} (4.30B) **Mean:** 7.77×10^{10} (77.7B)	Measures government spending on public goods, administration, healthcare, education, and social protection.	High positive skewness with mean **18x the median**. Shows that public sector financial capacity is tightly bound to the overall national economic scale.

Final Consumption Expenditure	**Extreme Outlier Intensity** Range: ~0 to ~1.95×10^{13}	**Median:** 2.17×10^{10} (21.7B) **Mean:** 3.26×10^{11} (326B)	Aggregates both private household and public government consumption spending combined.	Combines the skew of household and government expenditure. The upper tail extends to nearly 20 Trillion, reflecting the dominating weight of major consumer economies.
Gross Fixed Capital Formation	**Extreme Outlier Intensity** Range: ~0 to ~7.5×10^{12}	**Median:** 6.03×10^{9} (6.03B) **Mean:** 1.19×10^{11} (119B)	Measures net additions to fixed assets (infrastructure, machinery, equipment, land development).	Highlights global infrastructure investment concentration. The mean is **~20x higher** than the median, driven by heavy fixed-asset investment in top-tier developing and developed economies.

Gross Capital Formation	**Extreme Outlier Intensity** Range: ~0 to ~7.6×10^{12}	**Median:** 6.38×10^{9} (6.38B) **Mean:** 1.22×10^{11} (122B)	Total national investment including fixed asset formation plus net inventory accumulation.	Closely matches Gross Fixed Capital Formation with a slightly elevated baseline, showing that raw inventory variations add minor variance relative to heavy fixed capital investments.
Changes in Inventories	**Moderate-High Outlier Intensity** Range: -0.2×10^{11} to 1.75×10^{11}	**Median:** 3.43×10^{8} (343M) **Mean:** 4.08×10^{9} (4.08B)	Measures net annual changes in stocks of raw materials, work-in-progress, and finished goods held by businesses.	Unique bi-directional behavior: contains negative values representing inventory drawdowns/destocking. Outliers (> 100B) reflect large industrial stockpiling or supply chain buffering in dominant manufacturing economies.

Agriculture Hunting Forestry Fishing	**Extreme Outlier Intensity** Range: ~0 to ~1.35 × 10^{12}	**Median:** 2.66 × 10^{9} (2.66B) **Mean:** 1.99 × 10^{10} (19.9B)	Measures economic output and net value added from primary sector activities.	Severe right-skew with the mean **~7.5x higher** than the median. Over 190 countries cluster below 50B, while a few large agricultural nations push the upper tail past 1.3 Trillion.
Per Capita GNI	**High Outlier Intensity** Range: ~0 to ~235,000	**Median:** 6,755.00 **Mean:** 18,080.51	Measures average gross national income earned per person across economies.	Significant positive skew where the mean is **~2.7x higher** than the median. The majority of observations cluster under 20,000, while hyper-wealthy/tax-ha ven economies pull the extreme tail up to 235k+.

Population	**Extreme Outlier Intensity** Range: ~0 to ~1.43 × 10^9	**Median:** 6,792,908.5 (6.79M) **Mean:** 37,185,528.15 (37.19M)	Represents total resident population counts across individual countries/regions.	Extreme concentration in the baseline bin (< 50M) with a mean **~5.5x higher** than the median. Upper outliers near 1.4 Billion highlight global demographic megastates (e.g., China, India).
IMF Based Exchange Rate	**Extreme Outlier Intensity** Range: ~0 to ~42,000	**Median:** 8.72 **Mean:** 966.54	Measures official local currency units per US Dollar (or SDR base unit) defined by the IMF.	Massive positive skew with mean **~111x higher** than the median. Over 180 countries have exchange rates under 2,000, while hyper-inflated currencies form a long sparse right tail up to 40,000+.

AMA Exchange Rate	**Extreme Outlier Intensity** Range: ~0 to ~110,000	**Median:** 8.72 **Mean:** 1,318.12	Measures exchange rates calculated using UN Analysis of Main Aggregates adjustment methodology.	Matches the median of the IMF rate but exhibits even higher extreme outliers (up to 110,000). The mean is **~151x higher** than the median due to severe currency devaluation distortions in specific tail nations.

Box plots graphs inference or insights:-

Metric	Outlier Intensity & Range	Box Profile (IQR & Median)	Primary What It Indicates	Key Analytical Inferences
Imports Per Person	Extremely Heavy Right Tail (Outliers stretch up to ~235,000)	Compressed box near zero; median sits far to the left within the Interquartile Range (IQR).	High dispersion with numerous upper outliers beyond the 1.5 IQR boundary.	1. Import spending is heavily skewed by ultra-wealthy, import-reliant microstates and high-income nations. 2. Non-parametric methods or log-transformations are required to prevent outlier distortion.

Exports Per Person	Extremely Heavy Right Tail (Outliers stretch up to ~280,000)	Tightly compressed box near the lower bound; tight IQR.	Severe upper-end dispersion with a long chain of positive outliers.	1. Global trade production is concentrated in a tiny cluster of export powerhouses and oil/commodity-rich economies. 2. The lower 75% of nations operate within a very narrow, low-export band.
Investment Per Person	Heavy Right Tail (Outliers stretch up to ~58,000)	Narrow IQR compressed below 5,000; upper whisker terminates around 12,000.	Capital investment per capita shows high positive skewness and extreme high-end outliers.	1. Per-capita capital accumulation is minimal for the majority of nations, but reaches massive levels in advanced economies. 2. Outliers distort standard linear regressions unless treated.
GDP Per Person	Heavy Right Tail (Outliers stretch up to ~235,000)	Box is compressed toward the bottom end (<20,000); upper whisker stops near 50,000.	Clear structural ceiling for 75%+ of nations, with significant high-income outliers.	1. Visually reinforces the global economic divide: most nations cluster in the low-to-middle income box. 2. Extreme outliers require robust scaling or rank-based modeling.
Agriculture Share	Moderate Upper Outliers (4 distinct upper outliers between 0.43--0.67)	Broad IQR (0.02 -0.17); median sits near 0.07. Upper whisker reaches 0.38.	The main distribution is low and right-skewed, but bounded below 1.0.	1. High agricultural dependency (>40%) is confined to a tiny group of agrarian, developing economies.

				2. Most countries have successfully transitioned away from primary agricultural production.
Services Share	Minimal Upper Outliers (3 minor upper outliers between 0.70--0.76)	Wide, well-centered IQR spanning roughly 0.26 – 0.43; median near 0.33.	High symmetry across the central 50% of data with low outlier prevalence.	1. Services form a stable structural backbone across almost all national economies globally. 2. Standard parametric assumptions (e.g., normal distribution) hold well for this variable.
Household Share	Symmetric with Mild Bipolar Outliers (1 low outlier near 0.10; 3 high outliers between 1.08--1.40)	Centered, wide box (0.51--0.73); median sits near 0.62. Whiskers cover 0.20--1.06.	Well-balanced overall distribution with rare macro-accounting anomalies at the extremes.	1. Consumption accounts for a predictable 50%--73% chunk of GDP in almost every economy. 2. Outliers >1.0 indicate nations relying heavily on foreign aid or debt to fund domestic consumption.
Government Share	Moderate Upper Outliers (Outliers extend from 0.38 to over 0.90)	Compact IQR between 0.12--0.22; median near 0.17. Upper whisker ends near 0.33.	High cluster around 12%--22%, with a thin tail of high-spending states.	1. State intervention remains relatively bounded globally, but extreme outliers signal heavy state-run or welfare-heavy regimes. 2. Median is a highly representative measure for typical state involvement.
Investment Ratio	Low Outlier Density (3 upper outliers	Balanced IQR (0.18--0.28); median near 0.23. Whiskers	Symmetric core distribution with very few high-investment	1. Capital formation as a percentage of GDP is highly standardized across global economies (23%

	between 0.45--0.57)	span 0.03--0.43.	exceptions.	median). 2. Ideal candidate feature for modeling without requiring heavy pre-processing.
Trade Openness	Moderate High-End Outliers (Outliers cluster between 1.9 and 4.0)	Moderately wide box (0.56--1.10); median near 0.80. Upper whisker extends to 1.88.	Skewed upward by re-export hubs and hyper-open micro-economies.	1. Most economies trade between 56%--110% of their total GDP. 2. Outliers above 2.0 represent specialized international trade/financial hubs (e.g., Singapore, Hong Kong).

Import Dependency	Low-to-Moderate Upper Outliers (4 distinct upper outliers stretching between ~1.25 and 2.00)	Moderate IQR (approx. 0.30–0.60); median sits near 0.42. Whiskers extend from ~0.05 to 1.04.	Centered distribution with a moderate spread, bounded above 1.0 by specialized economies.	1. Most nations have an import dependency ratio between 0.30 and 0.60. 2. Outliers above 1.0 indicate highly open economies whose import values exceed domestic production output.
Export Dependency	Moderate-to-Heavy Upper Outliers (A sequence of ~10 upper outliers spanning from ~1.02 up to ~2.12)	Compact box positioned lower (IQR ~0.22–0.54); median near 0.34. Upper whisker terminates near 0.94.	Marked right-skewness with a longer trail of high-end trade outliers compared to imports.	1. The broad base of countries maintains low-to-moderate export reliance (< 0.54). 2. Extreme positive outliers pinpoint major global

				export/transshipment hubs and resource-rich supply nations.
Gross Domestic Product (GDP)	Extreme Right-Tail Outliers (Massive cluster up to ~0.5 10^13, with two mega-outliers at ~1.77 times 10^13 and ~2.33 times 10^13)	Highly compressed box near zero; median and IQR are squeezed tightly against the baseline.	Extreme macro-economic inequality and massive right-skewness on an absolute scale.	1. The vast majority of world economies sit in a low-GDP tier, while top global superpowers dwarf the median. 2. Strongly demands logarithmic scaling (log_10) for linear parametric modeling or regression tasks.
Gross National Income (GNI)	Extreme Right-Tail Outliers (Dense chain of outliers up to ~0.5 times 10^13, plus two distant points near ~1.76 times 10^13 and ~2.34 times 10^13)	Extremely compressed IQR near the vertical axis; upper whisker barely extends beyond zero.	Dynamic mirror profile to total GDP, demonstrating extreme wealth concentration.	1. Tracks nearly identically with GDP, confirming national income distribution is dominated by the top handful of economies. 2. Non-parametric methods or robust scaling are strictly required due to extreme leverage points.
Wholesale, Retail Trade, Restaurants & Hotels	Severe Upper-End Outliers (Cluster up to ~0.45 times 10^12, with extreme outliers at ~0.73 times 10^12, ~2.0 times 10^12, and ~3.52 times 10^12)	Box is heavily compressed near 0; lower and median boundaries collapse against the bottom axis.	Sizable scale disparity in domestic trade and consumer service sector volumes globally.	1. Absolute consumer trade and hospitality value is heavily driven by large population/high-income nations. 2. Typical economies operate at a fraction of the upper extreme's sector size.

Transport, Storage & Communication	Severe Upper-End Outliers (Outliers extend across ~0.1 --0.5 times 10^12, with mega-outliers near ~1.41 times 10^12 and ~2.48 times 10^12)	Highly compressed IQR near 0; upper whisker ends well below 0.1 times 10^12.	Distribution is severely right-skewed with a few massive logistics/teleco m markets globally.	1. Infrastructure and transport valuation tracks total economic size rather than broad global parity. 2. Outliers reflect large continental infrastructure networks and major telecom centers.
Total Value Added	Extreme Right-Tail Outliers (Outliers span up to ~0.5 times 10^13, with two extreme upper points at ~1.76 times 10^13 and ~2.32 times 10^13)	Box is squeezed paper-thin against 0; median and IQR boundaries are practically visually merged.	Complete structural alignment with GDP/GNI outlier distribution.	1. Re-affirms that productive economy value-addition is concentrated in a tiny tier of global economic giants. 2. Standard variance measures (e.g., standard deviation) will be heavily distorted by top outliers.
Mining, Manufacturing, Utilities	Extreme High-End Outliers (Cluster up to ~1.15 times 10^12, with severe standalone outliers at ~3.27 times 10^12 and ~5.78 times 10^12)	Heavily compressed box near zero baseline; upper whisker ends under 0.15 times 10^12.	Dominance of a few industrial behemoths in energy, mining, and industrial output.	1. Secondary/industrial sector value added is negligible for most nations compared to global manufacturing hubs. 2. The top outlier (~5.78) represents a massive global industrial manufacturing baseline.
Other Activities	Extreme High-End Outliers (Dense outlier chain up to ~0.22 times	Tightly compressed box near 0; median and central 50% sit very low.	High positive skewness driven by massive public administration, healthcare, or	1. Reflects broad economic scale where non-industrial service outputs scale with overall national wealth.

	10^13, with distant outliers at ~0.60 times 10^13 and ~1.28 times 10^13)		financial services sectors in top states.	2. Far-right outliers reflect high-income states with comprehensive, large-scale service sectors.
Manufacturing	Extreme High-End Outliers (Cluster extending to ~1.0 times 10^12, with extreme outliers near ~2.5 times 10^12 and ~4.88 times 10^12)	Compressed box hugging the zero axis; upper whisker terminates below 0.1 times 10^12.	Massive concentration of global manufacturing output in a few dominant industrial economies.	1. Highlights the disparity in global industrial production capacity. 2. Outliers require heavy feature transformation (e.g., log or Box-Cox transformation) before inclusion in standard machine learning models.

Imports Of Goods And Services	Heavy Upper Outliers (Dense chain up to 0.95 times 10^12, with major isolated outliers at 1.78 times 10^12, 3.07 times 10^12 and 3.40 times 10^12)	Compressed box hugging zero baseline (IQR ends below 0.10 times 10^12); upper whisker reaches 0.15 times 10^12.	High positive skewness in raw trade spending across global economies.	1. Most countries maintain relatively small overall import markets. 2. Severe upper outliers represent massive national import markets and key global trading economies.
Household Consumption Expenditure	Extreme High-End Outliers (Cluster extending to 0.27 times 10^13, with extreme individual outliers at 0.68 times 10^13 and	Box is squeezed paper-thin against 0; median and IQR boundaries are visually collapsed against the lower axis.	Immense wealth and aggregate consumer demand concentration globally.	1. Baseline consumer expenditure for typical economies is negligible compared to top-tier economies. 2. Extreme right-tail points require

	1.59 times 10^13)			non-parametric statistical methods or log-transformation.
Gross Fixed Capital Formation	Extreme High-End Outliers (Dense points up to 1.25 times 10^12, with two far-right outliers near 4.95 times 10^12 and 7.43 times 10^12)	Highly compressed box near zero; upper whisker terminates well under 0.2 times 10^12.	Severe geographical concentration of fixed asset investments (infrastructure, property, machinery).	1. The broad base of nations has limited absolute fixed capital formation. 2. Far-right extreme outliers signify massive domestic investment spending in dominant economies.
Gross Capital Formation	Extreme High-End Outliers (Cluster up to 1.25 \times 10^12, plus standalone mega-outliers at 4.93 times 10^12 and 7.58 times 10^12)	Paper-thin box hugging zero baseline; lower whisker and median bound tightly to the origin.	Mirrors Gross Fixed Capital Formation, confirming total national investment distribution is heavily skewed.	1. Reflects strong macro-structural correlation with fixed capital trends. 2. Parametric models will suffer severe leverage distortions without scaling.
General Government Final Consumption Expenditure	Severe Upper Outliers (Chain spanning up to 1.07 times 10^12, with distant outliers near 2.83 times 10^12 and 3.35 10^12)	The box is heavily squeezed near 0; The upper whisker ends below 0.1 times 10^12.	Scale disparity in public sector spending, state budgets, and government services globally.	1. Public sector spending is concentrated in high-GDP/large-population states. 2. The top two outliers reflect exceptionally high national public budget allocations.
Final Consumption Expenditure	Extreme High-End Outliers (Cluster up to 0.37	Compressed box hugging zero; upper whisker barely extends beyond	Aggregated national spending (household + government) is	1. Total consumption volume is anchored by a few huge consumer markets.

	times 10^13, with severe standalone outliers at 0.96 times 10^13 and 1.93 times 10^13)	0.05 times 10^13.	extremely right-skewed.	2. Requires robust statistical methods to handle influential points during modeling.
Exports Of Goods And Services	Heavy Upper Outliers (Dense chain up to 0.92 10^12, with extreme standalone outliers at 2.00 times 10^12, 2.55 times 10^12, and 3.53 times 10^12)	Compact box near baseline (IQR ends near 0.08 times 10^12); upper whisker extends to 0.15 times 10^12.	Severe right-skewness showing global trade export dominance among top economies.	1. The baseline distribution shows low-to-moderate export volume for most nations. 2. Outliers capture leading manufacturing export giants and resource-rich trade hubs.
Construction	Extreme High-End Outliers (Point cluster extending to 0.28 times 10^12, with massive upper outliers near 0.95 times 10^12 and 1.24 times 10^12)	Box compressed tightly near 0; upper whisker terminates below 0.03 times 10^12.	Highly uneven global distribution of construction sector GDP contribution.	1. Real estate and infrastructure construction activity is concentrated in a tiny subset of large economies. 2. Significant gaps between the main cluster and top outliers show major industrial disparities.
Agriculture, Hunting, Forestry, Fishing	Extreme High-End Outliers (Points extending up to 0.21 times 10^12, with isolated mega-outliers at 0.54 times 10^12 and 1.35 times 10^12)	Box squeezed against the zero baseline; upper whisker terminates under 0.02 times 10^12.	Concentration of primary sector agricultural output in giant agrarian economies.	1. Primary economic output is modest in absolute monetary value for most countries. 2. Top outliers represent nations with massive agricultural landmasses and outputs.

Changes In Inventories	Two-Sided Outliers (Both Negative and Positive) (Lower outliers down to 0.21 times 10^11; upper outliers stretching up to 1.71 times 10^11)	Narrow central box around 0 (IQR 0.00 – 0.03 times 10^11); whiskers span -0.03 times 10^11 to 0.05 times 10^11.	Dynamic variable reflecting stock buildup vs. drawdowns; unique among macroeconomic indicators for negative values.	1. Most countries maintain near-zero inventory adjustments in a given cycle. 2. Negative outliers reflect economic inventory drawdowns; high positive outliers show massive national stockpiling.

Per Capita GNI	Moderate-to-Heavy Upper Outliers (Continuous chain from 52,000 to 95,000, with major isolated outliers near 112,000, 198,000, and 234,000)	Wide box relative to axis scale (IQR approx. 3,000--21,000); median sits near 7,000. Upper whisker reaches 48,000.	High global inequality in individual earning power, though less concentrated at zero than absolute macro totals.	1. Most countries maintain a Per Capita GNI below 21,000. 2. Extreme upper outliers highlight specialized high-income economies, tax havens, or small population resource-rich nations.
Population	Extreme Upper Mega-Outliers (Continuous cluster up to 0.34 times 10^9 [340], plus two distinct mega-outliers at 1.40 times 10^9 and 1.43 times 10^9)	Paper-thin box hugging zero baseline (IQR ends below 0.03 times 10^9 [30]); upper whisker reaches 0.06 times 10^9.	Immense demographic concentration where a vast majority of nations have small-to-mediu m populations.	1. Typical countries have fewer than 30 million inhabitants. 2. The two far-right outliers represent demographic giants (China and India), which heavily distort sample means if unweighted.
IMF Based Exchange Rate	Severe Right-Tail Outliers	Extremely squeezed box near zero baseline; upper	Significant nominal currency denomination	1. Most national currencies trade at low nominal exchange rate figures relative to standard baseline

	(Points spread across 1,000 --14,000, with far-end outliers near 23,000,24,500, and 42,000)	whisker terminates below 500.	disparities and hyper-denominated national currencies globally.	units (e.g., SDR/USD). 2. Far-right outliers reflect economies with highly inflated or heavily devalued currency units.
AMA Exchange Rate	Extreme High-End Outliers (Cluster extending up to 14,000, with secondary outliers near 23,000--24,000, and a massive mega-outlier above 110,000)	Box is squeezed completely flat against zero; median and IQR are visually indistinguishable from the baseline.	Extreme nominal exchange rate variance under Alternative Conversion Factors (AMA).	1. Average market exchange rates are low for standard nations, but currency reform/inflation histories create severe leverage points. 2. The single far-right outlier (>110,000) represents a nation with extreme nominal currency devaluation that strictly requires log transformation.

Bivariate Graphs:-

[Correlation Graphs (scatterplot) - table with interpretation]

Graph Metric	Exports vs GDP	Trade Openness vs GDP	Imports vs GDP	Investment vs GDP	Population vs GDP	GDP vs GNI
Variables	X: Exports (1e12 USD)Y: GDP (1e13 USD)	X: Trade Openness (GDPExports+Imports ratio)Y: GDP (1e13 USD)	X: Imports (1e12 USD)Y: GDP (1e13 USD)	X: Investment (1e12 USD)Y: GDP (1e13 USD)	X: Population (1e9 people)Y: GDP (1e13 USD)	X: GDP (1e13 USD)Y: GNI (1e13 USD)
Correlation Value (r)	+0.875(Strong Positive)	−0.131(Weak Negative / Negligible)	+0.934(Very Strong Positive)	+0.941(Very Strong Positive)	+0.601(Moderate Positive)	+1.000(Perfect Positive)
Trendline Direction	Upward-sloping linear regression line.	Downward-sloping linear regression line.	Upward-sloping linear regression line.	Upward-sloping linear regression line.	Upward-sloping linear regression line.	Perfect 45◦ upward diagonal line (y≈x).
Data Point Distribution	Dense cluster near origin (0,0); high-value global outliers far to the top-right.	Mega-economies cluster at low openness ratios (<0.5); high openness ratios (2.0–4.0+) sit strictly at low GDP levels.	Dense cluster near origin (0,0); US and China sit far out as major high-import anchors.	Cluster below 1T investment; distinct split between US (top outlier) and China (far-right outlier).	Most nations cluster below 0.35B population; China and India sit far to the right at ≈1.4B.	Almost every single point sits directly on top of the linear regression line across all scale levels.
Core Economic Relationship	Higher export volume strongly aligns with a larger national economy, as exports directly feed GDP.	Trade openness ratio has no meaningful linear link to total absolute dollar GDP.	Higher GDP nations consume significantly higher total foreign import volumes.	Capital investment (Gross Fixed Capital Formation) tracks almost directly with aggregate economic output.	Larger labor pools generally support larger aggregate GDPs, but productivity creates wide variance.	Domestic production (GDP) and resident income (GNI) are virtually identical in global aggregate totals.

Limitation 1: Scale / Structural Bias	Scale Effect: High correlation reflects total country size rather than true export reliance.	Large Country Effect: Massive nations have low trade ratios because their huge internal markets lower relative trade dependency.	Accounting Paradox: Imports are subtracted in GDP (GDP=C+I+G+X−M), yet correlate positively due to purchasing power.	Mutual Causality: Investment creates GDP growth, but higher GDP provides savings to invest more (Accelerator Effect).	Missing Per-Capita View: Aggregate totals ignore individual living standards and income distribution entirely.	Accounting Identity: GNI is built directly on top of GDP (GNI=GDP+ Net Foreign Income); the 1.0 fit is mathematical.
Limitation 2: Causality Dynamics	Reverse Causality: Rich, high-GDP nations have the advanced capacity needed to produce massive export volumes.	Ratio vs. Absolute Mismatch: Compares a proportional percentage ratio against total nominal dollar size.	Purchasing Power Drive: High GDP/income causes high imports, not the other way around.	Quality vs. Quantity: Measures total spent, ignoring capital efficiency (e.g., productive tech vs. empty real estate).	Productivity Variance: Huge labor forces with low per-worker output (e.g., India) generate smaller total GDPs than tech/capital-rich nations (e.g., US).	Hides Tax Havens/Multinationals: Multi-trillion scale masks major percentage gaps in FDI hubs like Ireland (GDP ≫ GNI).
Limitation 3: Data Measurement Flaws	Gross vs. Net: Measures gross exports (X) instead of net exports (X−M), overstating trade strength for deficit nations.	Outlier Skew: Re-export trade hubs (e.g., Singapore) pull openness ratios high without having multi-trillion dollar GDPs.	Extreme Leverage: Large import-deficit nations (e.g., USA) heavily leverage and anchor the regression slope.	Model Divergence: Anchored by two opposing economic models (US consumption-led vs. China investment-led).	Demographic Composition Ignored: Fails to account for age distribution, working-age ratio, or workforce skills/education.	No Per-Capita View: Fails to show individual wealth, living standards, or local purchasing power.

When performing bivariate analysis on macro-level national economic data, the patterns across these six graphs reveal four fundamental takeaways:

1. The "Country Size" Scale Effect Dominates Nominal Relationships
Most macro-level economic variables (exports, imports, investment, GDP, and GNI) correlate very strongly with one another ($r > 0.87$). However, this is largely driven by absolute country scale rather than policy or performance. Large nations simply produce, invest, export, and import in massive dollar amounts. Without normalizing variables—such as converting values to **per-capita metrics** (GDP per capita) or **proportional ratios** (Investment as a % of GDP)—bivariate scatter plots primarily measure how big a nation is rather than how efficiently its economy functions.

2. High Correlation Does Not Prove Causation
A strong correlation coefficient (r) can easily mislead if taken at face value:

- **Imports vs. GDP ($r = 0.934$):** Imports are mathematically subtracted from GDP in national accounting ($GDP = C + I + G + X - M$), yet they show an extremely strong positive correlation because high domestic income and purchasing power drive import demand.

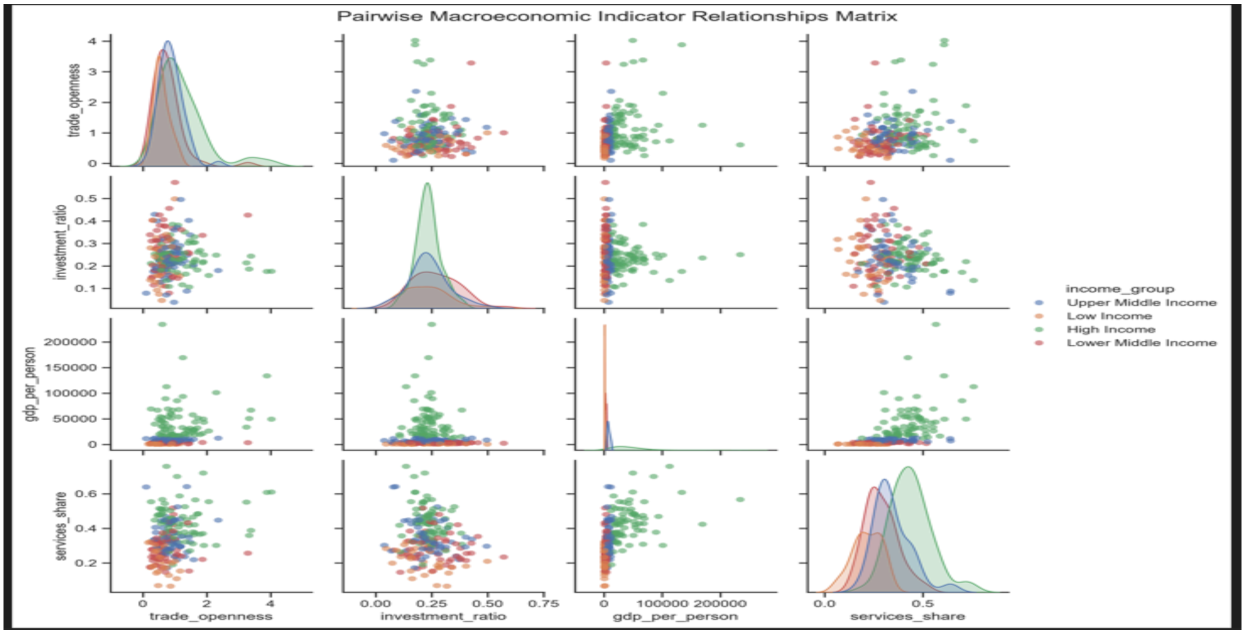
- **GDP vs. GNI ($r = 1.00$):** A perfect 1.0 correlation is an accounting identity—because GNI is calculated directly from GDP—rather than an empirical economic discovery.

3. Aggregate Measures Completely Obscure Individual Well-Being
Bivariate comparisons of aggregate nominal totals say almost nothing about living standards or economic development. A country can have a multi-trillion-dollar economy due to a massive population (e.g., India) or heavy capital investment (e.g., China), but still have a lower per-capita income and living standard than a smaller, highly productive economy with a modest total GDP.

4. Ratios and Rates Require Context
As seen in the **Trade Openness vs. GDP** graph ($r = -0.131$), pairing a percentage ratio against an absolute nominal total creates structural distortions. Large economies like the US naturally have lower trade-to-GDP ratios because of their massive domestic internal markets, whereas small global trade hubs (like Singapore or Luxembourg) have ratios well above 100% (1.0). Evaluating relative metrics against absolute totals can falsely imply negative or non-existent relationships.

Takeaway: Bivariate analysis is a useful starting point for visual inspection, but nominal macroeconomic variables must always be normalized, evaluated for direction of causality, and backed by multivariate regression models to draw valid policy or analytical conclusions.

[Pairplots - graph analysis]



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Here is a straightforward, section-by-section explanation of what each scatter plot in the pairplot matrix is showing.

Row 1: Trade Openness against Other Variables

Plot (1,2) – Trade Openness vs Investment Ratio

- Pattern:
  Most data points form a dense cluster between trade openness values of approximately 0.5–1.5 and investment ratios of about 0.15–0.35 across all income groups.
- Meaning:
  The amount a country trades relative to its GDP has very little relationship with the proportion of its GDP that is invested in capital formation. Countries with similar investment ratios can have very different levels of trade openness.

Plot (1,3) – Trade Openness vs GDP per Person

- Pattern:
  Lower-income and middle-income countries are concentrated near the bottom of the

graph but are spread across a wide range of trade openness values. High-income countries extend upward on the graph, with a few extreme observations exhibiting very high trade openness (greater than 3.0–4.0, or over 300–400% of GDP).

- Meaning:
  Greater trade openness does not automatically translate into higher income levels. However, extremely trade-open economies are typically wealthy global trade or financial hubs, such as Singapore or Luxembourg.

Plot (1,4) – Trade Openness vs Services Share

- Pattern:
  The scatterplot shows a loose upward-sloping pattern. Lower-income countries tend to have smaller service sectors and relatively low-to-moderate trade openness, while higher-income countries generally have larger service sectors and higher trade openness.
- Meaning:
  Countries with larger service sectors tend to be somewhat more integrated into international trade, although the relationship is only moderate.

Row 2: Investment Ratio against Other Variables

Plot (2,1) – Investment Ratio vs Trade Openness

- Pattern:
  This graph is simply the mirror image of Plot (1,2). It again shows no strong relationship between investment ratios and trade openness.
- Meaning:
  Countries that invest a larger share of GDP do not necessarily trade more relative to their economic size.

Plot (2,3) – Investment Ratio vs GDP per Person

- Pattern:
  Most observations form a narrow vertical band around investment ratios of approximately 20%–30% of GDP, regardless of income level.
- Meaning:
  Countries across all stages of development tend to invest a similar proportion of GDP into capital formation. High investment ratios are observed both in developing countries

building infrastructure and in developed countries investing in technology and productivity improvements.

Plot (2,4) – Investment Ratio vs Services Share

- Pattern:
  The points form a broad, scattered cloud with no obvious upward or downward trend.
- Meaning:
  The size of a country's service sector does not appear to determine the proportion of GDP that is invested in capital goods or infrastructure.

Row 3: GDP per Person against Other Variables

Plots (3,1) and (3,2)

- Pattern:
  These are mirror images of Plots (1,3) and (2,3). Most developing countries are concentrated near the lower end of the income scale, while a relatively small number of high-income countries extend far upward, producing a strongly right-skewed distribution.
- Meaning:
  GDP per person varies enormously across countries. Most nations have relatively modest income levels, while only a small number of countries achieve very high incomes.

Plot (3,4) – GDP per Person vs Services Share

- Pattern:
  This graph displays a clear positive relationship. Lower-income countries generally have service sectors contributing less than 40% of GDP, whereas high-income countries tend to have service sectors accounting for 50–70% or more of GDP.
- Meaning:
  This represents the strongest structural relationship in the entire pairplot. As countries become wealthier, their economies gradually transition away from agriculture and manufacturing toward service-based activities. This pattern reflects the process of structural transformation associated with economic development.

Row 4: Services Share against Other Variables

Plots (4,1), (4,2), and (4,3)

- Pattern:
  These are mirror images of Plots (1,4), (2,4), and (3,4). They consistently show that countries with larger service sectors tend to belong to higher-income groups.
- Meaning:
  The share of services in GDP distinguishes different stages of economic development much more clearly than trade openness or investment ratios. Countries with larger service sectors are generally more economically developed, while countries with smaller service sectors are more likely to be developing or lower-income economies.

[Pearson Correlation Heatmap Analysis]

The Pearson Correlation Heatmap is a correlation matrix that displays the strength and direction of the linear relationship between every pair of macroeconomic variables in the dataset.

Each cell contains the Pearson correlation coefficient, which ranges from -1 to +1.

- A value close to +1 indicates a strong positive relationship.
- A value close to 0 indicates little or no linear relationship.
- A value close to -1 indicates a strong negative relationship.

The heatmap allows us to quickly identify variables that move together, variables that move in opposite directions, and groups of variables that share similar economic characteristics.

Color Scale

The colours represent the strength and direction of the correlation.

- Off-White / Light Orange (approximately +0.8 to +1.0)
  - Indicates a very strong positive correlation.
  - Both variables increase together.
- Red / Magenta (approximately 0.0 to +0.4)
  - Indicates a weak to moderate positive correlation.
  - The variables move together only slightly.
- Dark Purple / Black (approximately -0.2 to -0.5)

- Indicates a negative correlation.
- As one variable increases, the other tends to decrease.

Key Structural Clusters in the Heatmap

1. The Large Light Central Square (Nominal Macroeconomic Aggregates)

Variables Included

- Construction
- Exports of Goods and Services
- Final Consumption Expenditure
- Gross Capital Formation
- Imports of Goods and Services
- Manufacturing
- Gross National Income (GNI)
- Gross Domestic Product (GDP)
- Other similar total-value macroeconomic indicators

Pattern

The centre of the heatmap forms a large off-white block where most correlations are very close to +1.0.

Meaning

These variables are measured as total monetary values.

Large economies naturally have:

- higher GDP,
- greater consumption,
- larger manufacturing sectors,
- higher investment,
- larger imports,
- and larger exports.

Consequently, these variables all increase together, producing extremely high positive correlations.

This pattern primarily reflects the **country size effect**, where larger economies have higher values across nearly all aggregate macroeconomic indicators.

2. The Per Capita Block (Bottom-Right Corner)

Variables Included

- GDP per Person
- Investment per Person
- Exports per Person
- Imports per Person

Pattern

A separate light orange block appears in the lower-right section of the heatmap with correlations generally ranging between approximately +0.7 and +0.9.

Meaning

After adjusting for population size, wealthier countries tend to exhibit:

- higher GDP per person,
- greater investment per person,
- higher exports per person,
- and higher imports per person.

Unlike total monetary values, these indicators reflect living standards and economic productivity rather than the absolute size of the economy.

3. The Negative Structural Relationships (Dark Purple / Black Areas)

Agriculture Share vs Services Share and GDP per Person

Pattern

Dark purple or black cells indicate moderate negative correlations, generally ranging between approximately -0.4 and -0.5.

Meaning

As countries become wealthier and GDP per person increases, the contribution of agriculture to GDP generally declines, while the services sector expands.

This reflects the process of **structural transformation**, where economies gradually transition from agriculture to manufacturing and eventually to service-based activities as they develop.

Trade Openness / Export Dependency vs Nominal Economic Totals

Pattern

Trade openness and export dependency appear as darker horizontal and vertical bands when compared with the large monetary aggregates.

Meaning

Trade openness and export dependency are ratio-based measures rather than absolute monetary values.

As a result, they do not increase simply because a country's economy becomes larger.

A country can have:

- a very large GDP with relatively low trade openness, or
- a relatively small GDP with extremely high trade openness.

Therefore, these variables show relatively weak correlations with the large aggregate economic indicators.

4. Year and CountryID

Pattern

The variables Year and CountryID appear as nearly white horizontal and vertical bands throughout the heatmap.

Meaning

These variables serve primarily as identifiers rather than economic indicators.

- Year identifies the time period of each observation.
- CountryID uniquely identifies each country.

Since they do not directly measure economic performance, they exhibit little or no meaningful linear relationship with the macroeconomic variables included in the analysis.

Key Takeaways from the Heatmap

1. Strong Evidence of Multicollinearity

The heatmap clearly demonstrates that many aggregate macroeconomic variables are highly correlated with one another.

Variables such as GDP, GNI, total consumption, imports, exports, manufacturing, and gross capital formation all move together because they largely reflect the overall size of a country's economy.

Including several of these highly correlated variables simultaneously in a multiple linear regression model may lead to **multicollinearity**, making coefficient estimates unstable and difficult to interpret.

2. Importance of Normalized Indicators

Indicators that are normalized by population or expressed as ratios provide much more meaningful information about a country's economic structure.

Examples include:

- GDP per Person
- Investment per Person

- Exports per Person
- Imports per Person
- Trade Openness
- Export Dependency
- Agriculture Share
- Services Share

These variables allow meaningful comparisons between countries of different sizes by removing the influence of population and total economic scale.

The Pearson Correlation Heatmap reveals two important characteristics of the dataset. First, most aggregate macroeconomic indicators are **strongly positively correlated** because they primarily reflect the overall size of a country's economy, highlighting the presence of significant multicollinearity among total monetary variables. Second, normalized indicators such as **per capita measures, sectoral shares, and trade ratios** capture structural differences in economic development that are not simply driven by country size. Overall, the heatmap demonstrates the importance of distinguishing between absolute economic measures and normalized indicators when interpreting relationships and selecting variables for statistical modelling and economic analysis.

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Spearman Rank Correlation Heatmap Analysis

Pearson Correlation vs Spearman Correlation

Before interpreting the heatmap, it is important to understand the difference between Pearson and Spearman correlation.

Pearson Correlation

Pearson correlation measures the strength and direction of a **linear relationship** between two variables. It assumes that variables change proportionally and is highly sensitive to extreme values (outliers).

Spearman Correlation

Spearman correlation measures the strength and direction of a **monotonic relationship** between two variables by comparing their ranks rather than their actual values. It evaluates whether variables generally increase or decrease together, regardless of whether the relationship is perfectly linear. Because it uses ranks, Spearman correlation is much less affected by extreme outliers and skewed data distributions.

Graph Overview & Color Scale

What the Graph Represents

The Spearman Rank Correlation Heatmap displays the strength and direction of the monotonic relationships between all macroeconomic variables in the dataset.

Instead of comparing raw numerical values, it compares the ranking of countries for each variable. This provides a clearer picture of how countries are ordered relative to one another across different economic indicators.

Color Scale

The colours represent the strength and direction of the rank correlation.

- Off-White / Light Peach (approximately +0.75 to +1.00)
  - Indicates a very strong positive rank correlation.
  - Countries that rank high in one variable also tend to rank high in the other.
- Red / Dark Magenta (approximately 0.00 to +0.50)
  - Indicates a weak to moderate positive rank correlation.
  - The rankings move together, but not consistently.
- Very Dark Purple / Black (approximately -0.50 to -0.85)
  - Indicates a strong negative rank correlation.
  - Countries that rank high in one variable consistently rank low in the other.

Key Structural Clusters in the Heatmap

1. The Large Central Light Block (Nominal Macroeconomic Totals)

Variables Included

- Construction
- Exports of Goods and Services
- Final Consumption Expenditure
- Gross Capital Formation
- Imports of Goods and Services
- Manufacturing
- Gross National Income (GNI)
- Gross Domestic Product (GDP)
- Other aggregate macroeconomic indicators

Pattern

A large off-white block appears in the centre of the heatmap, with Spearman correlation coefficients generally ranging between approximately +0.95 and +1.00.

Meaning

Countries maintain almost identical rankings across these aggregate economic indicators.

For example, a country ranked among the highest in GDP is also likely to rank among the highest in:

- Gross National Income
- Manufacturing output
- Consumption expenditure
- Investment
- Imports
- Exports

This indicates that the relative ranking of countries remains highly consistent across total economic measures.

2. The Strengthened Per Capita Block (Bottom-Right Corner)

Variables Included

- GDP per Person
- Investment per Person
- Exports per Person
- Imports per Person

Pattern

The lower-right corner forms a bright off-white cluster with very strong Spearman correlations, generally between approximately +0.85 and +0.98.

This cluster appears even stronger than in the Pearson Correlation Heatmap.

Meaning

Countries that rank highly in GDP per person also tend to rank highly in:

- Investment per person
- Exports per person
- Imports per person

Using rankings removes the influence of exceptionally large economies and highlights the consistent ordering of countries based on living standards and productivity.

3. Strong Negative Structural Relationships

Agriculture Share vs GDP per Person, Per Capita GNI, and Services Share

Pattern

Very dark purple or black cells indicate strong negative rank correlations, generally ranging between approximately -0.75 and -0.85.

Meaning

Countries with higher rankings in GDP per person and national income consistently rank among the lowest in agriculture's share of GDP.

Conversely, countries with a larger agricultural sector generally rank lower in terms of income and economic development.

This provides strong evidence for the theory of **structural transformation**, in which economies gradually shift from agriculture toward manufacturing and eventually to service-based industries as they develop.

Trade Openness and Export Dependency vs Exchange Rates and Aggregate Economic Totals

Pattern

Dark purple horizontal and vertical bands appear between trade-related ratios and many of the large aggregate economic variables, with correlations generally ranging between approximately -0.50 and -0.70.

Meaning

Large economies generally rank lower in terms of trade openness because they rely more heavily on domestic production and consumption.

In contrast, smaller economies often rank much higher in trade openness and export dependency because international trade constitutes a much larger proportion of their GDP.

This demonstrates that trade openness reflects the structure of an economy rather than its overall size.

4. Year and CountryID

Pattern

The variables Year and CountryID appear as nearly white horizontal and vertical bands across the heatmap.

Meaning

These variables function primarily as identifiers rather than economic indicators.

- Year identifies the time period of each observation.
- CountryID uniquely identifies each country.

Since they do not measure economic performance directly, they exhibit little or no meaningful rank correlation with the other macroeconomic variables.

Spearman Correlation is Important

1. Reduced Influence of Outliers

Macroeconomic datasets often contain extreme observations, such as very large economies like the United States and China or very wealthy economies such as Luxembourg.

These observations can strongly influence Pearson correlation coefficients because Pearson uses actual numerical values.

Spearman correlation reduces this problem by comparing country rankings rather than raw values, providing a more robust measure of association that is less affected by extreme observations.

2. Clearer Structural Relationships

The Spearman Heatmap reveals much stronger structural relationships than the Pearson Heatmap.

In particular, the strong negative relationship between:

- Agriculture Share
- GDP per Person
- Per Capita GNI
- Services Share

becomes much more pronounced.

This strongly supports established theories of economic development, particularly the **Structural Transformation Model**, which states that economies become less dependent on agriculture and increasingly dominated by manufacturing and services as income levels rise.

The Spearman Rank Correlation Heatmap demonstrates that many macroeconomic variables exhibit strong monotonic relationships even when the relationships are not perfectly linear. Compared with the Pearson Correlation Heatmap, Spearman correlation provides a more robust assessment because it is less influenced by extreme values and skewed distributions commonly found in international macroeconomic data. The heatmap confirms that countries maintain highly consistent rankings across major economic indicators such as GDP, GNI, investment, imports, and exports, while also highlighting strong structural relationships between income levels, sectoral composition, and economic development. Overall, the Spearman analysis provides a clearer understanding of the relative economic positions of countries and offers stronger evidence for patterns predicted by economic development theory.

Group comparisons .py graphs generated

[graphs inside country counts file]

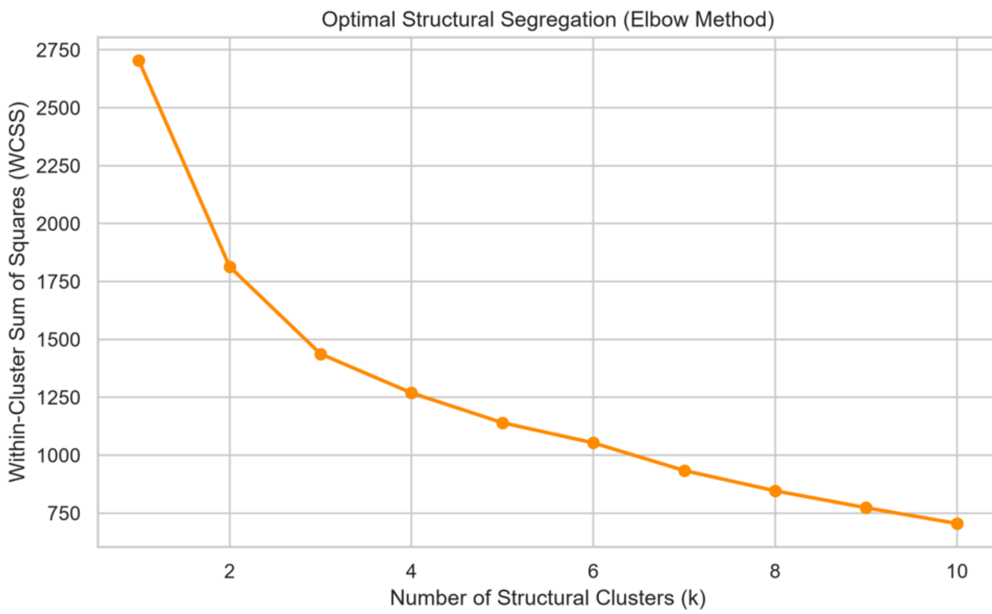
Graph Title	Categories & Counts	Primary What It Indicates	Key Analytical Insights
Number of Countries in each Development Status	• Developing: 106 • Developed: 75 • Under-developed: 31	Categorizes the total dataset of 212 countries into three classic macro-development stages.	1. Majority Non-Developed: Developing nations form the single largest category (106 out of 212, or exactly 50%), showing the dataset is heavily weighted toward emerging economies. 2. Asymmetric Distribution: Under-developed nations represent the smallest cohort (31 countries), indicating that fewer nations sit at the absolute lowest economic tier compared to those actively developing or fully developed.

Number of Countries in each Income Group	• High Income: 75 • Upper Middle Income: 54 • Lower Middle Income: 52 • Low Income: 31	Groups countries according to standard World Bank income classifications.	1. High-Income Dominance: High Income is the largest individual group (75 countries, matching the exact number of Developed countries in the dataset). 2. Progressive Decline at the Bottom: While Upper-Middle (54) and Lower-Middle (52) are almost evenly split, Low Income sits significantly lower (31 countries), highlighting a distinct drop-off at the bottom economic boundary.
Number of Countries in each GDP Group	• Very High: 54 • Medium: 54 • Low: 52 • High: 52	Displays the distribution of countries binned across four GDP magnitude tiers.	1. Quartile Binning Discretization: The nearly uniform distribution (54, 54, 52, 52; Total = 212) indicates this variable was transformed using statistical quartiles (Q_1, Q_2, Q_3, Q_4). 2. Balanced Sample Sizes: Quartile binning ensures equal sample sizes across categories, which prevents class imbalance issues when performing downstream statistical comparisons (e.g., ANOVA tests).
Number of Countries in each Population Group	• Low: 54 • Medium: 54 • High: 52 • Very High: 52	Displays the distribution of countries binned across four population size tiers.	1. Quartile Binning Discretization: Like the GDP group, population counts are split almost identically (54, 54, 52, 52), confirming quantile-based categorization. 2. Controlled Comparative Baseline: Using equal-sized population bins prevents demographic mega-outliers (like China or India) from skewing group-wise analytical comparisons.

Multivariate Graphs
[graphs generated inside the multivariate . py file]

Graphs generated inside folder called clusters

Graph 1: Optimal Structural Segregation (Elbow Method)
What the Graph Represents



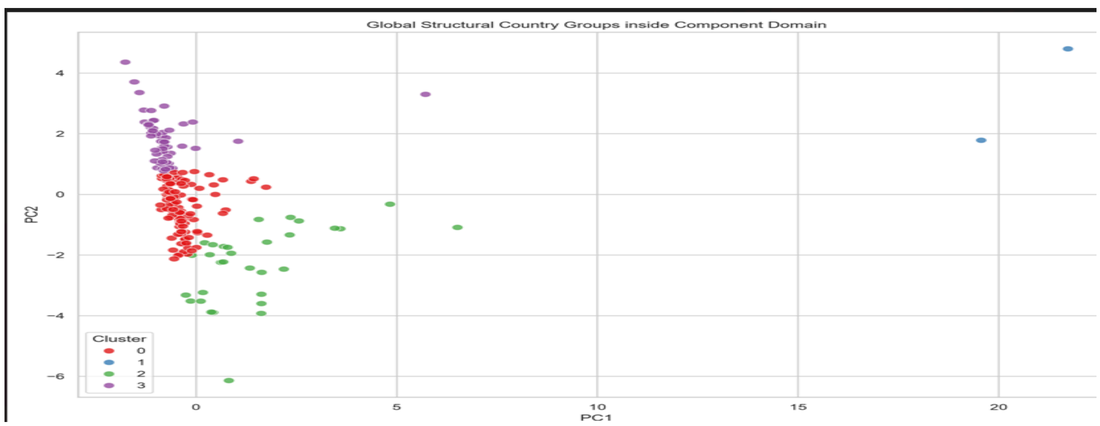
This graph illustrates the results of the Elbow Method analysis used to determine the optimal number of clusters (k) for partitioning the dataset via the K-Means algorithm. The horizontal axis measures the candidate number of structural clusters evaluated across a range from 1 to 10. The vertical axis measures the Within-Cluster Sum of Squares (WCSS), which quantifies the total variance or squared Euclidean distance between individual country data points and their respective assigned cluster centroids. Higher WCSS values indicate broad, heterogeneous groups, whereas lower values signify tightly bound, internally cohesive clusters.

How the Trend is Moving
The curve exhibits a sharp, downward trajectory beginning at k = 1 with a WCSS value exceeding 2,700, dropping rapidly to approximately 1,800 at k = 2 and around 1,430 at k = 3. This steep initial decline reflects significant structural separation as distinct economic tiers are isolated. Between k = 3 and k = 4, the rate of decrease noticeably decelerates, forming a distinct "elbow" inflection point. Beyond k = 4, the trend line transitions into a gradual, continuous slope, demonstrating diminishing marginal reductions in variance for every additional cluster added.

Inferences and Conclusions Regarding the Countries

The primary conclusion drawn from this trend is that global economies naturally segment into four major structural tiers rather than a continuous spectrum or a simple binary division. Selecting k = 4 captures the maximum meaningful structural variance among nations without introducing over-segmentation or noise. The sharp drop in early iterations indicates that countries possess distinct macroeconomic signatures that allow for clear grouping, while the flattening beyond four clusters confirms that adding further groups yields minimal additional clarity regarding country structural behavior.

Graph 2: Global Structural Country Groups inside Component Domain (PCA Cluster Plot)



What the Graph Represents

This scatter plot visualizes the four identified country clusters mapped within a reduced two-dimensional Principal Component Analysis (PCA) feature space. The horizontal axis, Principal Component 1 (PC1), captures the largest dimension of variance across the dataset, which corresponds primarily to absolute economic magnitude, industrial production, and total volume of trade. The vertical axis, Principal Component 2 (PC2), represents an orthogonal direction of variance driven by structural development metrics, intensity ratios, and per-capita indicators such as Per Capita GNI and currency valuation adjustments. Each data point represents an individual country, color-coded by its assigned cluster membership from 0 to 3.

How the Trend is Moving

The distribution of data points shows extreme spatial concentration along the lower-left region of the component space, paired with a long, sparse dispersion along both axes. The vast majority of nations cluster tightly near the origin where PC1 approx 0. As one moves rightward along PC1, the density of points drops precipitously, with only a tiny fraction extending past PC1 = 5 and reaching as far as PC1 approx 22. Vertical dispersion along PC2 is moderate near PC1 = 0, ranging from -6 to +4.5, but contracts as PC1 increases, showing that absolute economic scale eventually dominates structural variations.

Inferences and Conclusions Regarding the Countries

- **Cluster 0 (Red - Developing & Emerging Economies):** Occupies the dense region near the origin (PC1 approx 0, PC2 approx -2 to 0.8). This indicates that most of the world's nations share relatively small absolute market sizes and moderate per-capita indicators, placing them in a unified baseline developmental category.
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- **Cluster 1 (Blue - Global Economic Megastates):** Positioned at extreme positive values of PC1 (PC1 > 19). These outliers reflect economic super-powers (such as the United States and China) whose sheer scale in GDP, trade, and consumption sets them entirely apart from the rest of the global economy.
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- **Cluster 2 (Green - Mid-to-Large Industrial & Resource Nations):** Extends horizontally along positive PC1 (1 < PC1 < 7) with negative PC2 values down to -6. This captures major regional hubs, industrial economies, and resource-heavy nations (such as India or Brazil) that possess significant market volume but distinct structural signatures compared to high-income nations.
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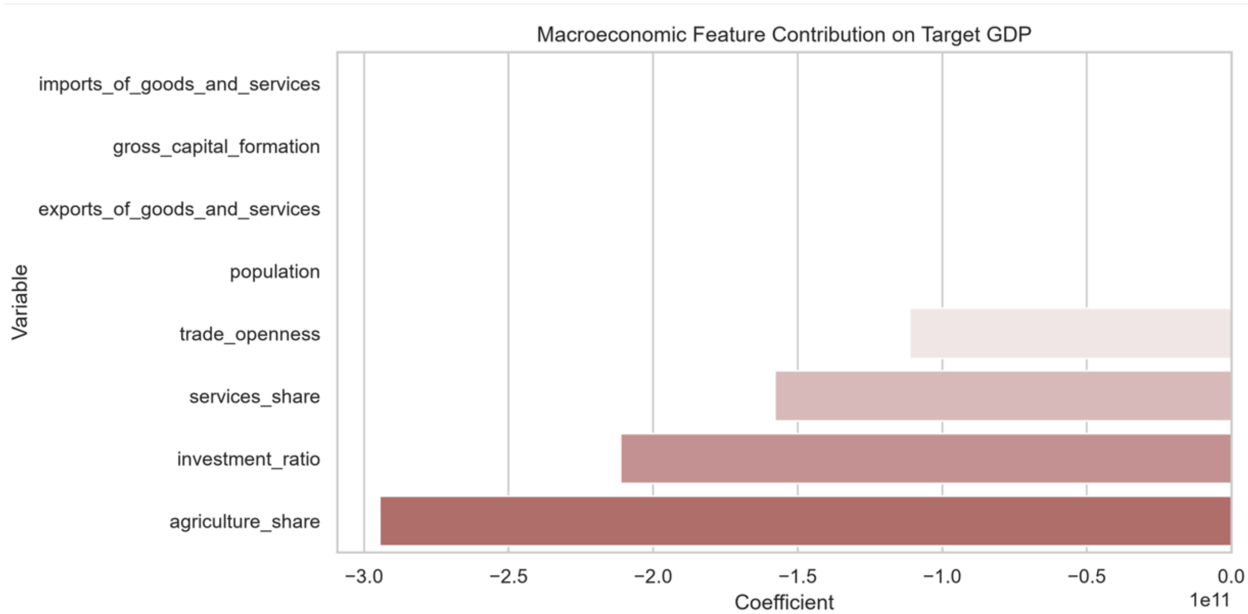
- **Cluster 3 (Purple - High Per-Capita & High-Value Economies):** Spans vertically along high positive PC2 (0.8 < PC2 < 4.5) while staying near low PC1 values. This highlights smaller nations and specialized economies (such as financial hubs or high-income microstates) that exhibit exceptionally high wealth per capita or strong service sectors despite having small absolute economic footprints.

REGRESSION ANALYSIS: MACROECONOMIC FEATURE CONTRIBUTION AND MULTIPLE REGRESSION PERFORMANCE

Macroeconomic Feature Contribution on Target GDP



What the Graph Represents

This horizontal bar chart illustrates the standardized regression coefficients of key macroeconomic variables used to model Target Gross Domestic Product (GDP). The vertical axis lists the predictor features evaluated, including economic sector shares (agriculture_share and services_share), ratio indicators (investment_ratio and trade_openness), demography (population), and aggregate trade and capital metrics (imports_of_goods_and_services, gross_capital_formation, and exports_of_goods_and_services). The horizontal axis displays the magnitude and sign of each regression coefficient, measured in scaled macroeconomic impact (×10^11). Bars extending to the left indicate negative structural relationships with overall GDP output.

How the Trend is Moving

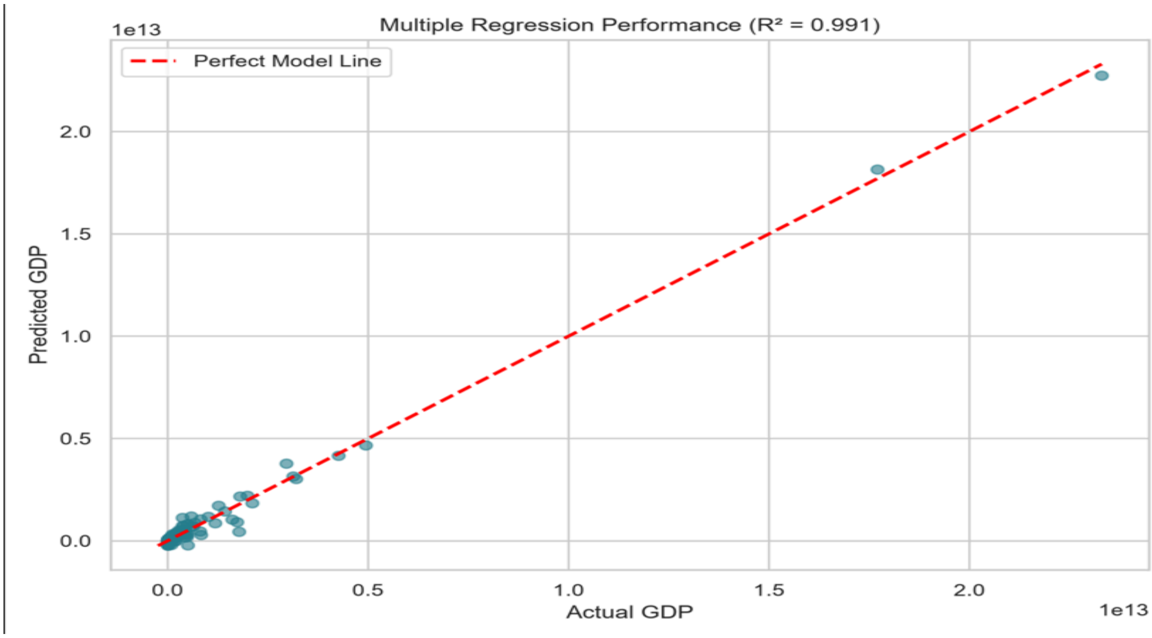
The graph exhibits a clear hierarchical pattern of variable impacts. Sectoral percentage shares and structural ratios display strong negative coefficients that systematically increase in magnitude as one moves down the chart. agriculture_share shows the largest negative coefficient, approaching -3.0 × 10^11, followed by investment_ratio at around -2.1 × 10^11, services_share at approximately -1.6 × 10^11, and trade_openness near -1.1 × 10^11.

Conversely, scale-based absolute variables, such as imports_of_goods_and_services, gross_capital_formation, exports_of_goods_and_services, and population, have coefficient values hovering near zero relative to the ratio variables.

Inferences and Conclusions Regarding the Countries:

- Agrarian and Primary Sector Nations (e.g., Developing Economies in Sub-Saharan Africa and South Asia):
  The strong negative coefficient for agriculture_share (approximately -2.9 × 10^11) reflects fundamental structural economic principles such as the Petty-Clark law. Countries where agriculture accounts for a high proportion of GDP consistently generate significantly lower total national GDP, indicating lower overall industrialization and labour productivity.
- Capital-Accumulating and Fast-Growing Emerging Markets:
  The negative coefficient for investment_ratio (approximately -2.1 × 10^11) highlights that developing and emerging economies typically spend a much larger percentage of their GDP on capital investment, such as infrastructure and basic machinery, relative to their existing small output base. Highly developed nations operate on vast established capital stocks where new gross investment represents a smaller percentage of total GDP.
- Transitioning and Service-Oriented Economies:
  A high relative share of baseline or informal services (services_share) yields a negative structural contribution relative to advanced industrial production, keeping predicted total GDP lower for transitioning nations.
- Small Open Economies vs. Sovereign Megastates:
  Small nations, such as trade hubs or microstates, exhibit exceptionally high trade_openness ratios (Trade-to-GDP percentage), but their absolute economic scale remains small, creating a negative relative coefficient. Conversely, absolute dollar-value trade variables, such as imports and exports, scale linearly with global economic giants, absorbing raw market volume while the ratio indicators capture the structural development stage.

Multiple Regression Performance (Actual vs. Predicted GDP)



What the Graph Represents

This scatter plot assesses the predictive accuracy of the multiple linear regression model by comparing observed national values (Actual GDP on the horizontal axis) against model estimations (Predicted GDP on the vertical axis), both scaled in tens of trillions of US Dollars ($\times 10^{13}$).

A dashed red diagonal reference line represents a theoretical "Perfect Model Line" ($y = x$), where predicted values exactly match actual values.

The chart header explicitly notes an exceptionally high Coefficient of Determination ($R^2 = 0.991$), indicating that the model explains 99.1% of the variance in global GDP.

How the Trend is Moving

The data points tightly align along the red 45-degree reference line across multiple orders of magnitude.

The vast majority of nations are clustered densely near the origin, below approximately 0.5×10^{13} or 5 trillion, reflecting the large number of small-to-midsize global economies.

A few isolated data points extend far along the diagonal line, reaching past 1.75×10^{13} and up to nearly 2.4×10^{13}, or approximately 24 trillion.

Residual dispersion around the line remains tight even in the high-value regions, indicating relatively consistent predictive performance across different economic scales.

Inferences and Conclusions Regarding the Countries:

- Global Economic Megastates (United States and China):
  These countries are positioned at the top-right extremity of the reference line, above approximately 1.75×10^{13} and extending toward 2.3×10^{13}. The model accurately predicts the extreme macroeconomic scale of the world's two largest economies without substantial underprediction or obvious non-linear distortion at the upper boundary. Their structural inputs, including large populations, high capital formation, and large trade volumes, align closely with their actual GDP values.
- Mid-to-Large Developed and Major Emerging Economies (e.g., Japan, Germany, India, UK, France, and Brazil):
  These countries are positioned along the middle segment of the line, approximately between 0.2×10^{13} and 0.5×10^{13}. Their observations fit closely to the ideal prediction line ($y = x$), indicating that their GDP values are well captured by the combination of industrial investment, service-sector activity, and population scale.
- Small-to-Mid Developed and Developing Nations (The Majority of Global Economies):
  Most observations are tightly clustered near the origin, approximately from 0 to 0.2×10^{13}. Their close grouping around the reference line suggests that the model performs well across smaller economies. Minor deviations near the origin may reflect structural differences, informal economic activity, or exchange-rate variations, although the overall residual variation remains small relative to the global variance, as reflected by the R^2 value of 0.991.